

## HERU PURNAMA

### Research and Development Engineer | Network & System Engineer

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herupurnama.com/HeruPurnama\_Portfolio.pdf

IT Infrastructure & R&D Engineer with strong experience in network engineering, system administration, automation, and cybersecurity, combined with hands-on experience in IoT and embedded system development. Skilled in building and integrating microcontroller-based systems, real-time sensor data pipelines, and secure communication between hardware and backend systems. Experienced in designing and optimizing enterprise infrastructure and automating workflows across multi-client environments to improve operational efficiency and reliability.

## EDUCATION

### Universitas Pendidikan Indonesia

#### Electrical Engineering

2019 – 2023

- Participated in Kampus Merdeka Program.
- Managed university server infrastructure (Proxmox, routers, switches, firewalls, NAT, port forwarding).
- Contributed to SmartVest and Smart Parking research projects.
- Developed Flask-based TOPSIS web application for solar panel research.
- Creating Electrical Planning and Room Drawing.

### SMKN 5 Bandung

#### Computer and Network Engineering

2016 – 2019

- Actively involved in IT Club organization.
- Experienced in various school projects involving configuration of MikroTik routers, 3D modeling using Blender, game development in Unity, creating virtual lab environments using Linux, and developing basic CRUD systems using PHP CodeIgniter.

## WORKING EXPERIENCE

### Information Technology

Cikarang

#### LGE Indonesia Research and Development Center

Dec 2025 – Present

- Managed lifecycle, compliance, and deployment strategy for 600+ enterprise devices, improving asset traceability and operational readiness.
- Administered and monitored 140+ CCTV units, improving surveillance reliability and incident response.
- Managed end-to-end procurement process for CCTV and access control systems, ensuring compliance and cost efficiency while supporting operational expansion.
- Conducted internal bidding and vendor selection to ensure cost efficiency and compliance.
- Maintained and monitored IT asset inventory to support operational needs.
- Implemented and managed security solutions such as Data Loss Prevention (DLP), Endpoint Security, Network Access Control (NAC), and Wireless ISP (WISP).
- Integrated Cloud Extreme Controller API with LGE internal portal to automate network management processes.
- Developed an automated monitoring system for 140+ CCTV and access control devices, reducing manual monitoring effort and improving incident response time.
- Controlling External Storage (Flash Drive, HDD, SSD) through full procurement lifecycle.
- Delivered cybersecurity awareness training to researchers and employees covering phishing, software supply chain risks, rogue access points, and social engineering threats.
- Managing employee onboarding and offboarding processes, including access provisioning and revocation.

### Network and System Engineer

Bandung

#### Teradata Megah

June 2024 – Dec 2025

Working at an IT solutions company serving the finance sector as part of a four-person team, supporting 40 internal users (including programmers and business support staff) and 6 banking clients. Coordinate with various partners, including ISPs and payment gateway providers, and responsible for managing the Network and System Infrastructure, ensuring the Data Centre operates reliably 24/7.

- Managing virtualization environments using VMware and Citrix.
- Implementing load balancing with Nginx and Citrix to optimize and distribute traffic efficiently.
- Database Replication, to ensure high availability data.
- Automating IPsec and OpenVPN tunnels for secure, rapid site-to-site deployments, integrated with Policy-Based Routing (PBR) and Network Address Translation (NAT).
- Administering network devices including Checkpoint, Mikrotik, and Aruba for security, routing, and wireless connectivity.
- Operating Extended Detection Response: XDR Bitdefender, EDR Eset to enhance endpoint security.

## ACTIVITIES

### Scalable Web Service With GoLang

Hacktiv8

<https://www.herupurnama.com/projects/2-GoLang-Project>

Indonesia, Remote

Feb 2024 - Mar 2024

Built and deployed scalable web services using Go (Golang) as part of a backend development program. Key projects include:

- Developing code to import data from Excel.
- Creating a REST API, and designing a minimal web game similar to Fallout Shelter.
- Creating MyGram API, which includes user management (login and register), photo uploads, commenting functionality, and social media interactions (post, get, put, delete) related to userIDs, and deploying it to Railway.

### Artificial Intelligence Mastery Program

Orbit Future Academy

<https://www.herupurnama.com/shortcut/relate-project>

RELATE (Repair Electronic Website) – Group Project Lead

Indonesia, Remote

Feb 2022 - Jul 2022

A group project to develop a server-side rendered (SSR) web application that helps users identify their device's screen condition and receive service recommendations using custom-trained Machine Learning models. I led a cross-functional team with expertise in Machine Learning (ML), Natural Language Processing (NLP), Computer Vision (CV), and Data Science (DS), while also contributing as a Full Stack Developer using Python and Django. Key projects include

- Implemented Git and Trello to streamline team collaboration and agile workflow.
- Integrated an NLP chatbot that recommends relevant YouTube videos and Google Maps locations based on user queries.
- Integrated Computer Vision to classify smartphone screen damage into categories (e.g., minor, moderate, severe).
- Deployed the application on Heroku and a private server, supporting features like authentication, CRUD operations, Google APIs, and a user profile system.

### Computer Network Engineer & IT Helpdesk

DitSTI ITB

Bandung, Indonesia

Jul 2018 – Sep 2018

- Offer client services, including hardware upgrades and repairs (primarily laptops), installing, licensing, and repairing software to various operating systems.
- LAN installation in new building and configured server environment using ISPConfig.

## PROJECTS

### Smart Home Automation & SmartVest Telegram API Integration

<https://www.herupurnama.com/shortcut/smart-home>

Apr 2023 - May 2023

Developed an IoT monitoring system using ESP32 and MQTT for real-time data transmission. A Python Flask server acts as a middleware bridge between the SIM800L module and external services due to TLS version limitations on the SIM800L. The server receives sensor and GPS data through lightweight GET/POST API endpoints, processes the data, stores it in a Supabase database, and forwards notifications to Telegram via webhook integration.

Example API endpoint:

/api/koordinat?lat=x&long=y&battery=z&altitude=a

### Security Method Analysis on MikroTik RouterOS

<https://www.herupurnama.com/shortcut/security-analysis>

August 2023

Researched MikroTik RouterOS security filtering performance under DDoS attacks using TOPSIS and entropy-weight analysis. Compared filter drop, tarpit, reject, and raw drop methods based on CPU, memory, and traffic metrics, concluding that Raw Drop provided the most efficient mitigation performance.

## CONTRIBUTION

### Improve LGERC Workflow

Utilizing Python Selenium and UI Path to do website/ERP automation to input USB data to:

- Increased data input accuracy and consistency.
- Reduced repetitive manual tasks and processing time.
- Minimized human errors through automated validation and data handling.

### Improve Teradata Workflow & Disaster Recovery Process

- Identified deployment bottlenecks caused by repetitive manual connection configuration during application releases across 20+ banking client environments.
- Designed and implemented a reusable connection profile repository by maintaining validated conn.bin files for each client environment, eliminating repetitive use of the Connection Creator tool during deployments.
- Reduced deployment effort, configuration time, and human error across multiple production and staging application servers through standardized connection management.

- Proposed and developed a Proof of Concept (PoC) to streamline application deployment by minimizing manual front-end and back-end connection configuration activities.
- Designed a segregated Disaster Recovery testing architecture (DRCv2) to isolate Proof of Concept and disaster recovery validation activities from the primary Disaster Recovery Center (DRC) environment.
- Reduced the risk of data inconsistency between the Data Center (DC) and DRC during testing scenarios, eliminating the need for full database re-synchronization after DRC testing activities.
- Improved operational efficiency, deployment consistency, and disaster recovery readiness through process standardization and infrastructure optimization.

## VOLUNTEER EXPERIENCE

### KKN Community Service

SDN Cibanteng & SMPN 59

<https://www.herupurnama.com/shortcut/kkn>

Bandung, Indonesia

January 2023

Led a multidisciplinary team to promote digital literacy in the community, focusing on cybersecurity and the use of technology. A key highlight was my article on Kompasiana, titled "Cara Mengamankan Diri di Dunia Digital," which received over 28,989 views. I also took initiative to merge and design a collaborative book chapter involving contributions from 31 participants.

### Digital Literacy Program Highlights:

Research & Content Development:

- Compiling comprehensive references on Cyber Security and Semiconductor-War.
- Gathering information on account cracking techniques to better educate against them.
- Developing engaging Cyber Security socialization materials for students.

Planning for Technical Demonstrations:

- Designing and preparing a website functionality simulation to illustrate digital concepts.
- Developing and preparing materials for a smart home demonstration, showcasing how relays and dot matrices can be controlled.
- Assembling microcontroller equipment, relays, and dot matrices to support the smart home demonstration.
- Setting up equipment for running text displays.

## CERTIFICATION

### Information and Communication Technology (ICT)

- Mikrotik: MTCNA – Jan 2019
- Cisco: Network Automation Engineering Fundamentals Specialization, Endpoints and Systems (Coursera)- March 2024
- Cisco CyberOps Associate (CCOA) - May 2024
- JavaScript Beginner (Udemy and Skilvul)- March 2024

- Basic (Python, SQL), Intermediate (SQL) HackerRank- Feb 2024

- Pure Storage Field Engineer, FlashBlade and FlashArray English Course

- ETS TOEIC score 570- Feb 2019
- Centralcourse TOEFL score 553- Sep 2023

## SKILLS

I have broad experience across multiple IT fields, including system integration, troubleshooting, automation, and web technologies. I enjoy adapting to new challenges and building efficient solutions.

### Networking & Security:

- Check Point, MikroTik, Aruba, VPN/IPsec, NAC, DLP, Endpoint Security, Wireshark.

### Infrastructure and Devops:

- VMware, Citrix, Proxmox, Linux, Nginx, Database Replication, Git, MQTT, REST API, Supabase

### Programming & Automation:

- Python, Go, JavaScript, Flask, Django, Selenium, UiPath